NAVAL CANNON

CANNON WERE FIRST USED at sea in the 14th century, and evolved through wrought iron, cast bronze, and cast iron production methods. The very earliest of these were breech-loaders, but muzzle-loaders quickly took over and held sway for four centuries until the advent of shell-firing breech-loaders in the 19th century. Naval cannon varied enormously in terms of caliber and size, the largest being the "cannon royal" that fired a 66 lb shot, down to smaller guns such as the culverin, which had a shot of around 17 lb.



ROPE WADDING Coils of old rope were beaten into shape to make wadding. These were then used to hold powder and ball firmly in place in the barrel of the cannon.



SALT BOX This held two ready powder charges stored side by side, and had leather straps and fittings to prevent sparks. Salt inside the box kept the charges dry.



CARTRIDGE Fabric sheets were sewn into cylinders as containers for powder charges, each bag being filled with a precise amount of powder to suit a particular gun.



Matches were lit at the start of battles and then placed in the match tub

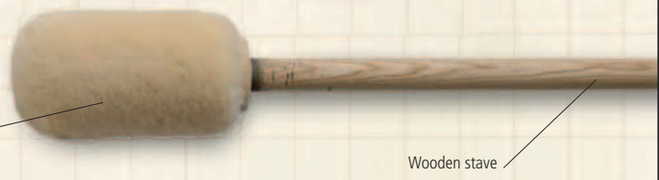
MATCH TUB This small wooden keg was used to hold slow matches, which were kept burning behind the gun, to ignite it if the gun lock failed in some way. Made from hemp, the matches were boiled in an alcohol and saltpeter solution, which caused the matches to burn slowly.



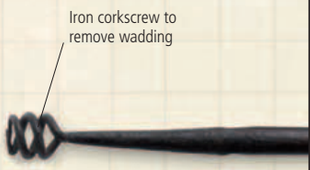
CASE OF WOOD These wooden cylinders were used for holding and carrying a single cartridge safely. As with gun carriages, elm was the preferred wood because of its anti-splinter properties.

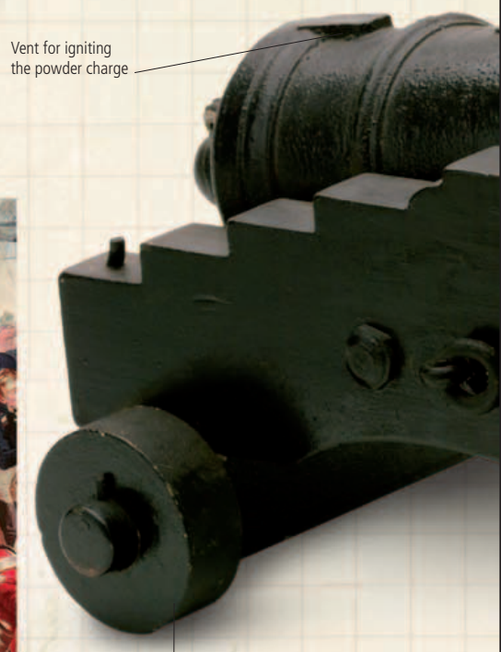












GUN DRILL

Naval cannon were usually mounted on wheeled wooden carriages, allowing them to run backward under recoil. The backward motion was controlled by restraining ropes running around the breech or the carriage. The ropes were critical—a 32-pounder gun could fly 50 ft (15 m) across a deck if allowed to freewheel. Carriages were often made of elm, which produced less splinters than other woods if struck by an enemy shell. Despite such precautions, being a gunner was an extremely dangerous job, with recoil and accidental explosions being just two of the hazards.

